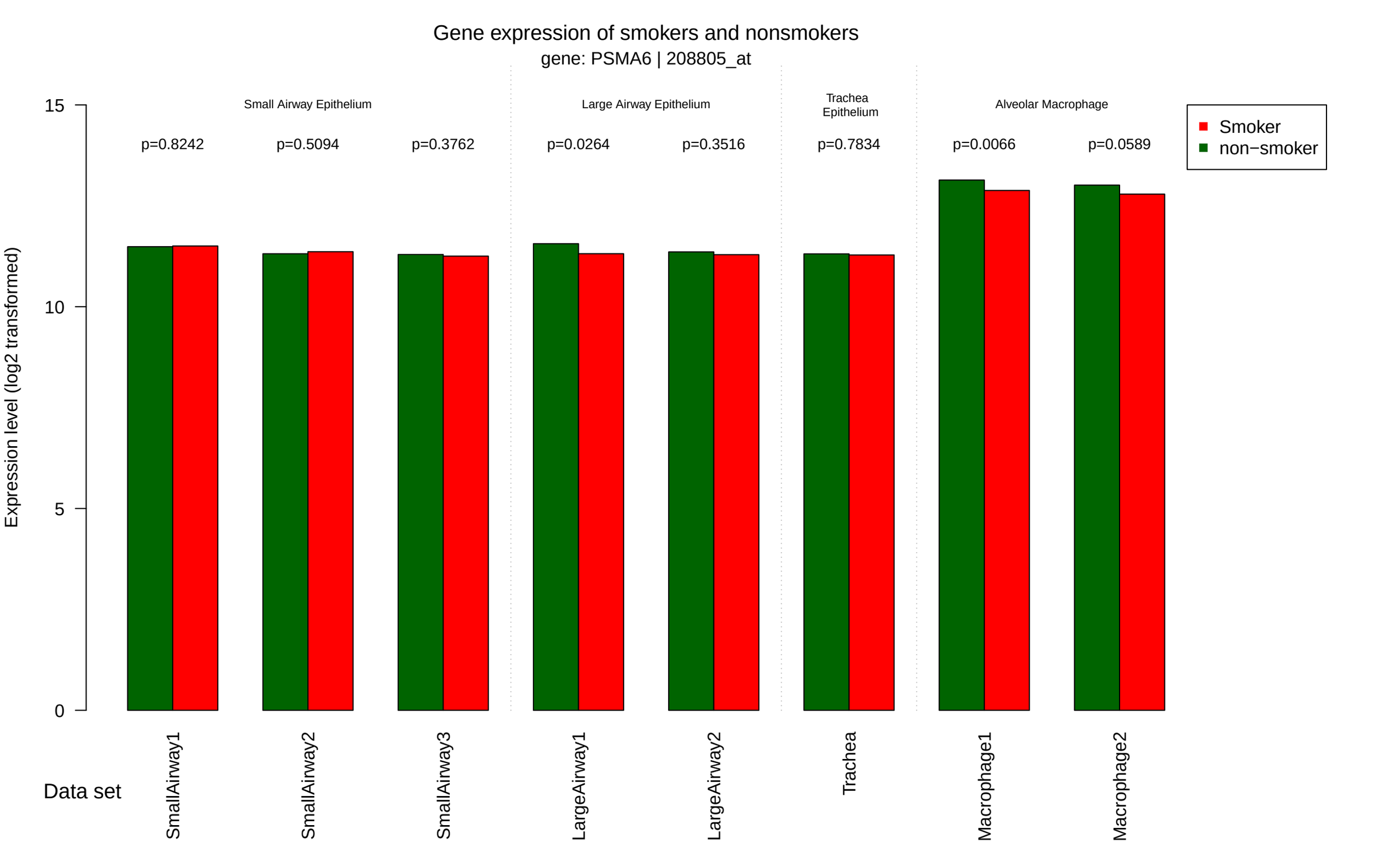
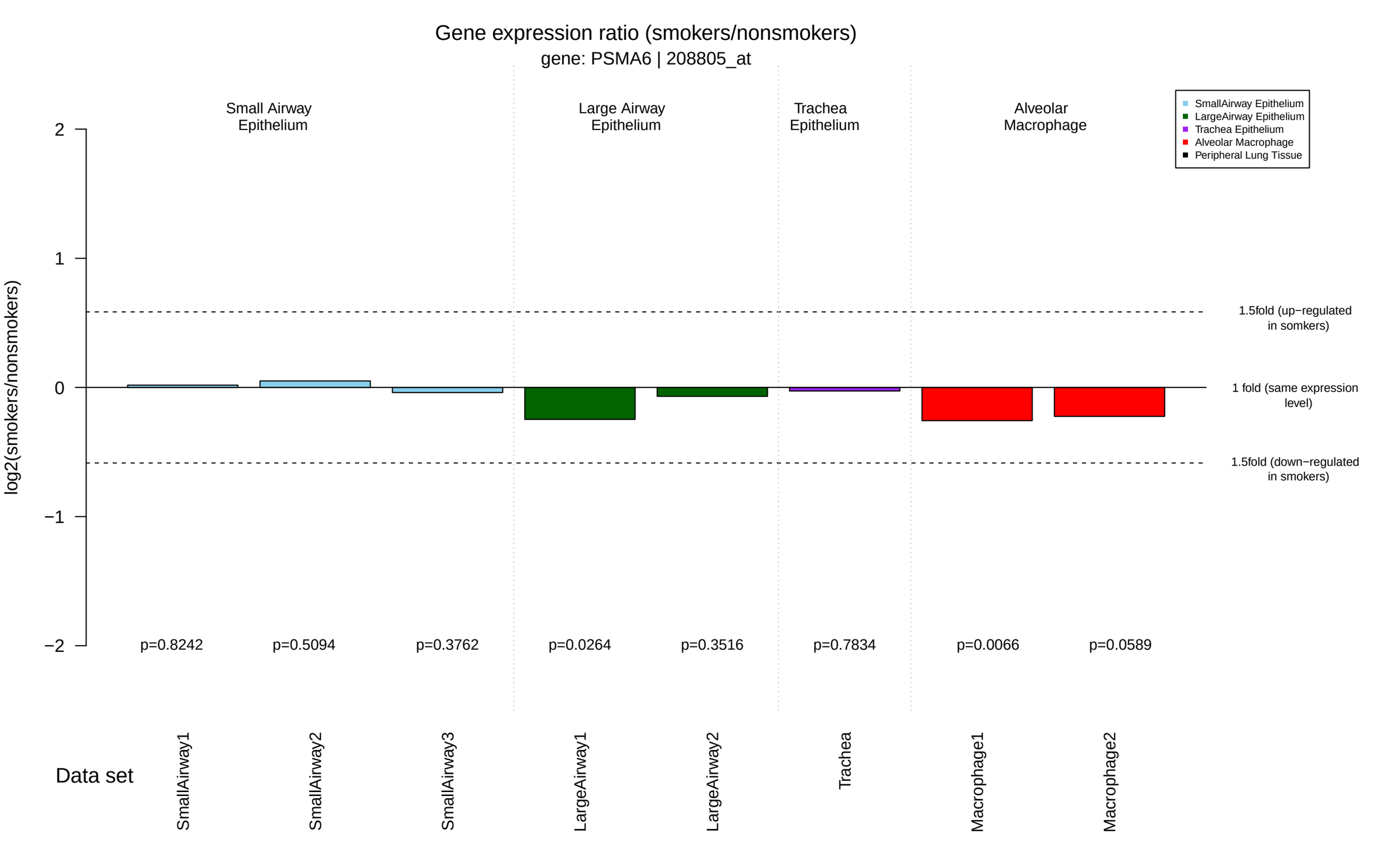
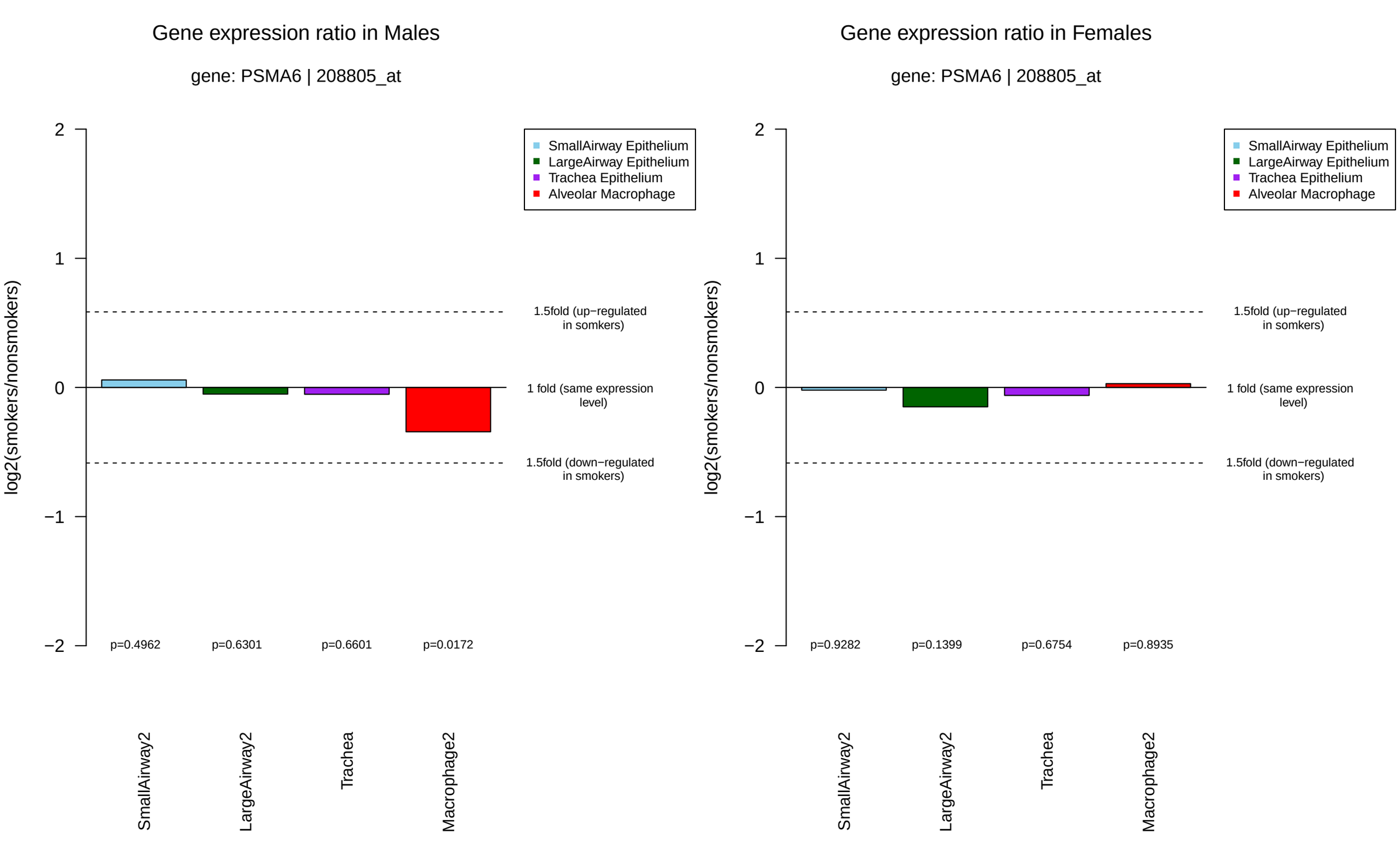
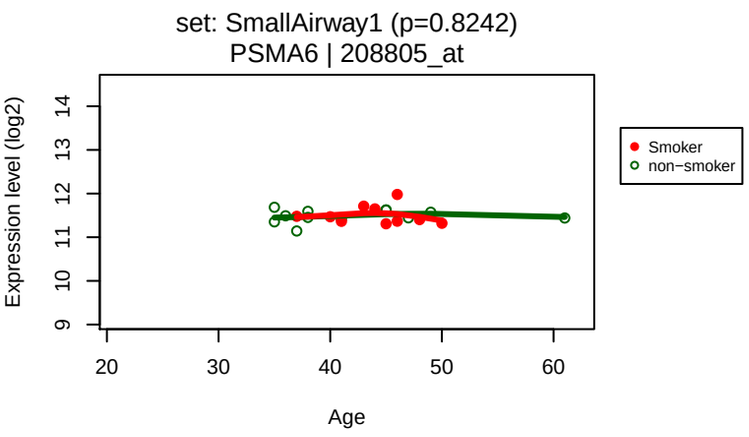
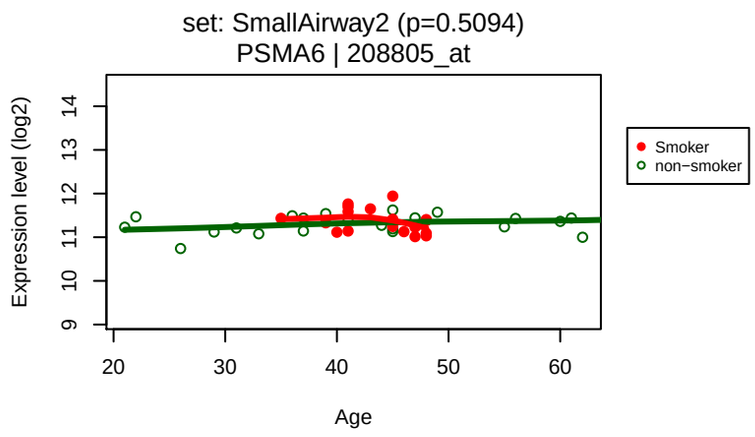
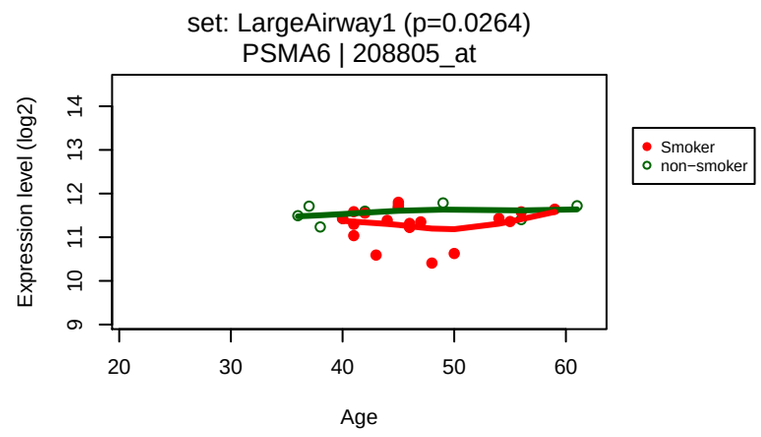

Gene expression of smokers and nonsmokers
gene: PSMA6 | 208805_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8242
p=0.5094
p=0.3762
p=0.0264
p=0.3516
p=0.7834
p=0.0066
p=0.0589
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

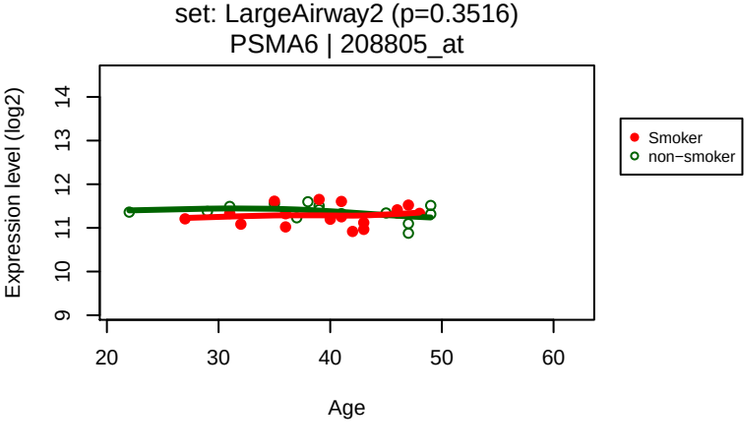

Gene expression ratio (smokers/nonsmokers)
gene: PSMA6 | 208805_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8242
p=0.5094
p=0.3762
p=0.0264
p=0.3516
p=0.7834
p=0.0066
p=0.0589
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

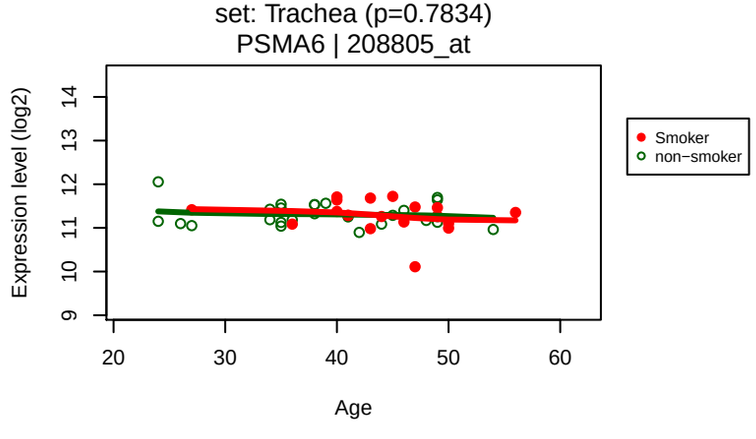

Gene expression ratio in Males
gene: PSMA6 | 208805_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4962
p=0.6301
p=0.6601
p=0.0172
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMA6 | 208805_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9282
p=0.1399
p=0.6754
p=0.8935
SmallAirway2
LargeAirway2
Trachea
Macrophage2

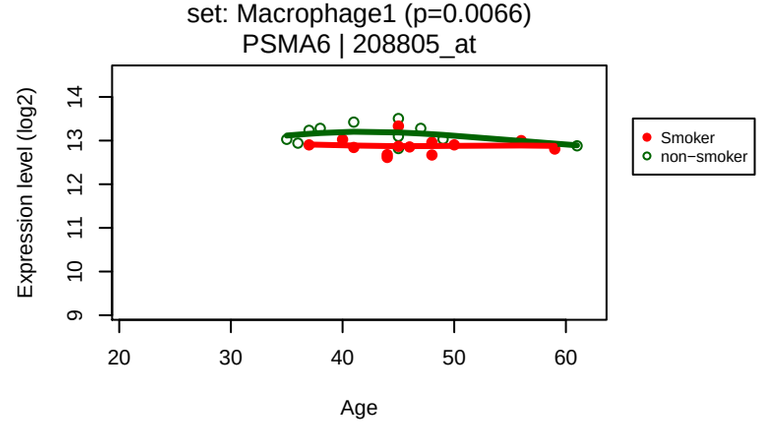

set: SmallAirway1 (p=0.8242)
PSMA6 | 208805_at
Expression level (log2)
Age
Smoker
non−smoker

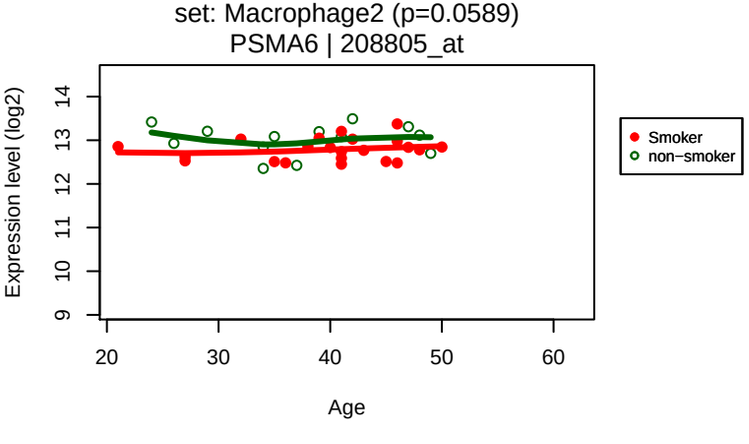

set: SmallAirway2 (p=0.5094)
PSMA6 | 208805_at
Expression level (log2)
Age
Smoker
non−smoker

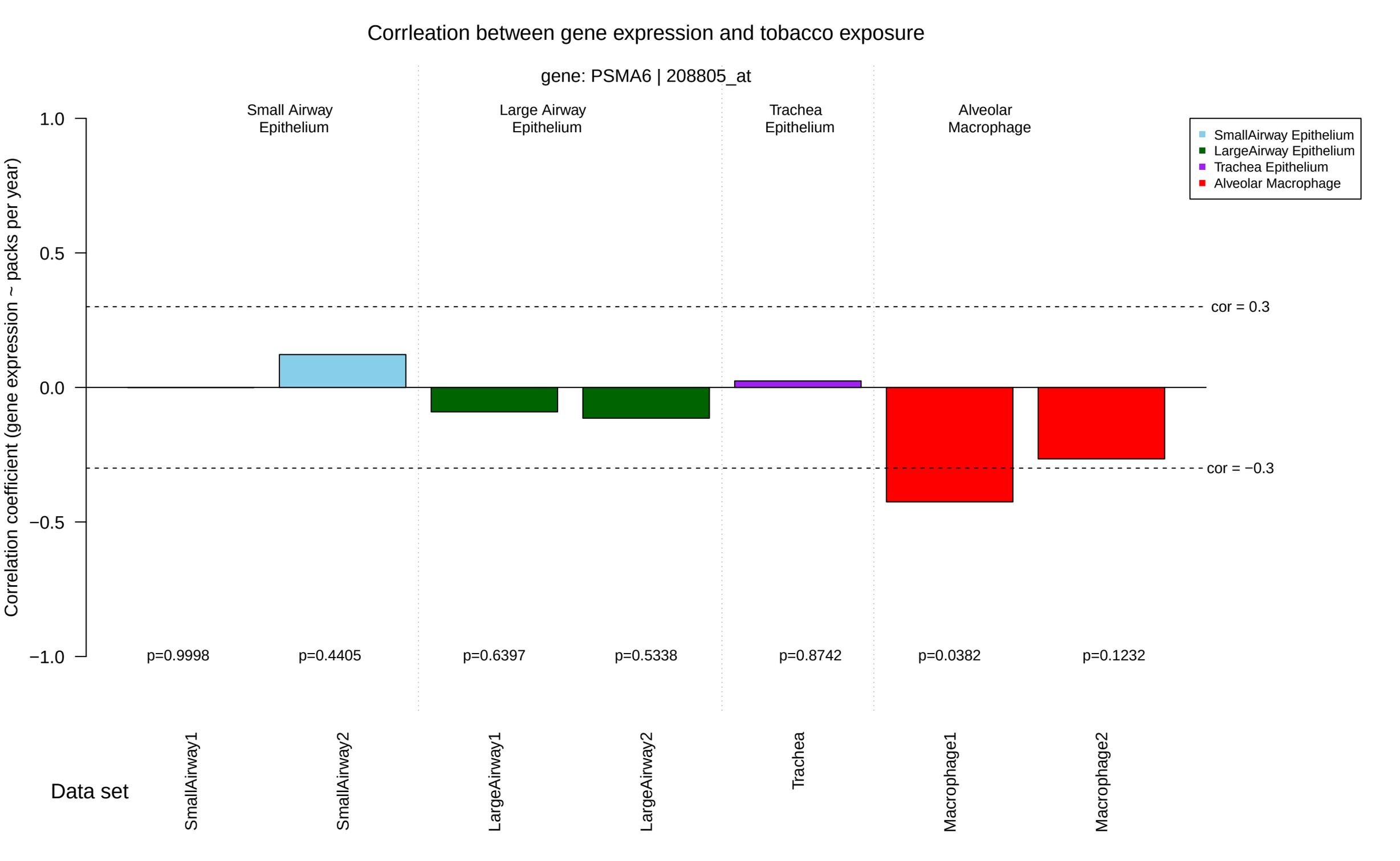

set: LargeAirway1 (p=0.0264)
PSMA6 | 208805_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3516)
PSMA6 | 208805_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7834)
PSMA6 | 208805_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0066)
PSMA6 | 208805_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0589)
PSMA6 | 208805_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMA6 | 208805_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9998
p=0.4405
p=0.6397
p=0.5338
p=0.8742
p=0.0382
p=0.1232
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2